Q1. Product Description for a Smart Fitness Watch

Aim: To design Zero-shot, One-shot, and Few-shot prompts to generate a product description for a Smart Fitness Watch.

Algorithm:

1. Start the program.
2. Define a task to generate a product description for a Smart Fitness Watch.
3. Create a Zero-shot prompt without examples.
4. Create a One-shot prompt with one example.
5. Create a Few-shot prompt with multiple examples.
6. Generate the product descriptions using each prompt.
7. Display and compare the descriptions.
8. Stop the program

1. Zero-shot Prompt

Prompt:

Write an attractive product description for a Smart Fitness Watch. Include its important features, benefits, and a call to action. Use simple and professional English.

Output:

Smart Fitness Watch – Your Everyday Fitness Partner

Stay healthy and connected with our Smart Fitness Watch! Track your daily steps, heart rate, sleep patterns, and workout activities with ease. Enjoy smart notifications, a stylish design, and comfortable wear throughout the day. Monitor your progress and stay motivated to achieve your fitness goals. Get your Smart Fitness Watch today and take the first step toward a healthier lifestyle!

2. One-shot Prompt

Prompt:

Example:

Product: Wireless Earbuds

Description: Enjoy clear sound, comfortable fitting, and long battery life with our wireless earbuds. Perfect for music, calls, and travel. Order yours today!

Now, write a similar product description for a Smart Fitness Watch. Include features, benefits, and a call to action.

Output:

Smart Fitness Watch – Smart Living, Better Fitness

Experience smarter fitness tracking with our Smart Fitness Watch! Monitor your heart rate, count your steps, track your sleep, and record your workouts. Stay connected with convenient smart notifications and enjoy its lightweight, modern design. Whether you are exercising or working, this watch helps you stay active and organized. Upgrade your lifestyle with a Smart Fitness Watch today!

3. Few-shot Prompt

Prompt:

Example 1:

Product: Backpack

Description: Carry your essentials comfortably with our lightweight and durable backpack. Perfect for school, travel, and everyday use. Shop now!

Example 2:

Product: Water Bottle

Description: Stay hydrated throughout the day with our reusable water bottle. Its portable design makes it ideal for workouts and travel. Get yours today!

Now, write a product description for a Smart Fitness Watch following the same style. Include features, benefits, and a call to action.

Output:

Smart Fitness Watch – Make Every Step Count

Transform your fitness routine with our Smart Fitness Watch! Track your steps, monitor your heart rate, analyze your sleep, and record your daily workouts. Its sleek design and smart notifications make it perfect for exercise, college, work, and everyday activities. Stay motivated,

monitor your progress, and work toward your health goals. Get your Smart Fitness Watch today and make every step count!

Result: Zero-shot prompting generates a description without examples, one-shot uses one example, and few-shot uses multiple examples to guide the output.

Q2. Blog on "Applications of Artificial Intelligence in Healthcare"

Aim: To design Zero-shot, One-shot, and Few-shot prompts to generate a 200-word blog about AI applications in healthcare.

Algorithm:

1. Start the program.
2. Define the topic as "Applications of Artificial Intelligence in Healthcare."
3. Create a Zero-shot prompt to generate a 200-word blog.
4. Create a One-shot prompt with one sample blog.
5. Create a Few-shot prompt with multiple sample topics and outputs.
6. Generate the blog using each prompting method.
7. Display the generated blogs and check their word counts.
8. Compare the structure, clarity, and completeness of the blogs.
9. Stop the program.

1. Zero-shot Prompt

Prompt:

Write a 200-word blog titled "Applications of Artificial Intelligence in Healthcare." Explain how AI is used in disease diagnosis, medical imaging, personalized treatment, drug discovery, and patient monitoring. Include benefits, challenges, and a conclusion. Use simple English.

Output:

Applications of Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is transforming healthcare by helping doctors make faster and more informed decisions. AI technologies analyze large amounts of medical data and support healthcare professionals in improving patient care.

One important application is medical imaging. AI systems help analyze X-rays, CT scans, and MRI images to identify possible abnormalities. AI can also support early disease detection by recognizing patterns in patient records.

Another application is personalized treatment. AI helps doctors study individual patient information and consider suitable treatment options. Healthcare chatbots provide basic information, appointment reminders, and answers to common questions.

AI also supports drug discovery by helping researchers analyze chemical compounds and identify potential medicines. Wearable devices use intelligent systems to monitor heart rate, sleep, and physical activity.

Despite these benefits, AI has several challenges. Patient privacy, cybersecurity, biased data, and inaccurate predictions must be addressed. AI systems require careful testing and human supervision to ensure safe use.

In conclusion, Artificial Intelligence offers valuable opportunities to improve diagnosis, treatment planning, medical research, and patient monitoring. When implemented responsibly, AI can support healthcare professionals, improve efficiency, and contribute to better patient outcomes. However, human expertise remains essential for safe and effective healthcare.

2. One-shot Prompt

Prompt:

Example:

Title: Applications of Technology in Education

Technology improves education through online learning, digital classrooms, personalized lessons, and educational applications. It makes learning more accessible and interactive. However, internet access and data privacy remain challenges. Responsible use of technology can improve educational experiences.

Now, write a 200-word blog titled “Applications of Artificial Intelligence in Healthcare.” Follow the example's simple, informative style. Cover applications, benefits, challenges, and a conclusion.

Output:

Applications of Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is becoming an important technology in modern healthcare. It helps doctors analyze information, identify possible diseases, and improve the delivery of medical services.

One major application of AI is medical image analysis. AI tools can examine X-rays and scans to identify patterns that require medical attention. These tools support doctors in detecting diseases earlier. AI can also analyze patient records to help identify health risks.

Personalized medicine is another important application. AI helps healthcare professionals study patient information and consider appropriate treatment options. Chatbots can answer common questions, send appointment reminders, and provide general health guidance.

AI also supports drug discovery by helping researchers study large datasets and identify promising compounds. Smartwatches and other wearable devices can monitor heart rate, sleep, and physical activity, helping users understand their health patterns.

Although AI offers many benefits, challenges remain. Patient privacy, data security, biased information, and incorrect predictions require attention. Medical professionals must supervise important decisions and verify AI-generated recommendations.

In conclusion, Artificial Intelligence can improve healthcare by supporting diagnosis, research, patient monitoring, and treatment planning. With responsible implementation and proper human supervision, AI can help healthcare professionals provide more efficient and informed patient care.

3. Few-shot Prompt

Prompt:

Example 1:

Topic: AI in Education

AI supports personalized learning, automated assessment, and intelligent tutoring. It helps students learn at their own pace. However, privacy concerns and unequal access must be addressed.

Example 2:

Topic: AI in Transportation

AI supports traffic prediction, route optimization, and autonomous driving. It can improve efficiency and safety, but reliability and security remain important concerns.

Now, write a 200-word blog titled “Applications of Artificial Intelligence in Healthcare.” Use a title, explain several applications, discuss benefits and challenges, and conclude with the importance of responsible AI.

Output:

Applications of Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is improving healthcare by supporting medical professionals in diagnosis, treatment planning, and patient monitoring. It can analyze large datasets and identify patterns that may be difficult to recognize manually.

One major application is medical imaging. AI systems assist doctors in examining X-rays, CT scans, and MRI images to identify possible abnormalities. Predictive analytics can also help identify patients who may need additional medical attention.

AI contributes to personalized medicine by helping professionals compare patient information and medical evidence. Healthcare chatbots can answer routine questions and provide appointment reminders. In pharmaceutical research, AI helps scientists analyze compounds and investigate potential medicines.

Wearable devices also use intelligent technologies to track heart rate, sleep, and physical activity. These features help people monitor their health and encourage healthy habits.

AI offers benefits such as improved efficiency, decision support, and faster data analysis. However, healthcare organizations must address privacy, cybersecurity, biased datasets, and unreliable predictions. AI tools need appropriate testing, and medical professionals must remain responsible for clinical decisions.

In conclusion, Artificial Intelligence has significant potential to improve healthcare services. Its applications range from medical imaging to drug discovery and remote monitoring. Combining reliable AI systems with professional medical expertise can support safer, more effective, and more accessible healthcare.

Result: The three prompting methods generate informative blogs. Examples in one-shot and few-shot prompts guide the structure, style, and content of the generated blog.

Q3. Article Summarization into 50 Words

Aim: To design three prompting methods to summarize an article into 50 words and compare the generated summaries based on accuracy, completeness, and readability.

Algorithm

1. Start the program.
2. Read the given article.
3. Create a Zero-shot prompt to summarize the article in 50 words.
4. Create a One-shot prompt with one example of article summarization.
5. Create a Few-shot prompt with multiple summarization examples.
6. Generate a summary using each prompt.
7. Count the words in each summary.
8. Compare the summaries based on accuracy, completeness, and readability.
9. Display the summaries and comparison results.
10. Stop the program.

Article

Title: Artificial Intelligence in Healthcare

Artificial Intelligence is increasingly used in healthcare. It helps doctors analyze medical images, predict health risks, support personalized treatment, and manage hospital workflows. AI-powered wearable devices monitor activity and vital signs. However, healthcare AI faces challenges such as data privacy, biased datasets, and inaccurate predictions. Human supervision and careful testing are essential for safe and responsible use.

1. Zero-shot Prompt

Prompt:

Summarize the following article in exactly 50 words. Preserve the main ideas, important applications, benefits, and challenges. Do not introduce information that is absent from the article.

Article: [Insert the article provided above]

Output — 50 words:

Artificial Intelligence supports healthcare through medical image analysis, health-risk prediction, personalized treatment, hospital workflow management, and wearable monitoring. These

applications can improve medical decision-making and patient care. However, privacy concerns, biased datasets, and inaccurate predictions remain challenges. Careful testing and human supervision are essential for safe and responsible healthcare AI implementation.

2. One-shot Prompt

Prompt:

Example:

Article: Solar energy produces electricity from sunlight. It is renewable and can reduce dependence on fossil fuels. However, installation costs and weather conditions can affect its use.

Summary: Solar energy generates renewable electricity, reduces reliance on fossil fuels, and offers environmental benefits. However, installation expenses and weather conditions can influence its effectiveness.

Now summarize the healthcare article in exactly 50 words. Preserve its main points and use clear English.

Article: [Insert the article provided above]

Output — 50 words:

AI is improving healthcare by assisting medical image analysis, predicting health risks, supporting personalized treatment, managing hospital workflows, and monitoring vital signs. Despite these benefits, privacy concerns, biased datasets, and inaccurate predictions remain important challenges. Careful testing and human supervision are necessary to ensure that AI is used safely and responsibly in healthcare.

3. Few-shot Prompt

Prompt:

Example 1:

Article: Online learning provides flexible access to educational resources. Students can learn remotely, but internet access and reduced face-to-face interaction may create challenges.

Summary: Online learning offers flexible remote education but may be limited by internet access and reduced face-to-face interaction.

Example 2:

Article: Electric vehicles reduce direct exhaust emissions and can lower running costs. However, charging infrastructure, battery production, and purchase prices remain concerns.

Summary: Electric vehicles can reduce exhaust emissions and running costs, although charging access, battery production, and purchase prices remain challenges.

Now summarize the healthcare article in exactly 50 words. Include the major applications, benefits, challenges, and need for human supervision.

Article: [Insert the article provided above]

Output — 50 words:

Healthcare AI supports medical imaging, health-risk prediction, personalized treatment, hospital management, and wearable monitoring. It can improve healthcare decision-making and patient services. However, privacy risks, biased datasets, and inaccurate predictions require attention. Careful testing and human supervision are essential to ensure that AI systems are used safely and responsibly in healthcare settings.

Comparison of summaries

Criteria	Zero-shot	One-shot	Few-shot
Accuracy	High	High	High
Completeness	Covers major points	Covers major points	Covers major points
Readability	Clear	Clear and concise	Clear and structured
Examples provided	None	One	Two

Result: All three summaries preserve the article's key ideas. One-shot and few-shot prompting provide examples that guide the summarization style.

Q4. Professional Email Requesting Leave Due to Illness

Aim: To design Zero-shot, One-shot, and Few-shot prompts to generate a professional leave-request email and compare the emails based on tone, grammar, formatting, and completeness.

Algorithm:

Start the program.

1. Define the task of writing a professional leave-request email due to illness.
2. Create a Zero-shot prompt without examples.
3. Create a One-shot prompt with one sample leave email.
4. Create a Few-shot prompt with multiple sample emails.
5. Generate a leave email using each prompting method.
6. Display the generated emails.
7. Compare the emails based on tone, grammar, formatting, and completeness.
8. Stop the program.

1. Zero-shot Prompt

Prompt:

Write a professional email to a college faculty member requesting leave due to illness. Include a suitable subject, polite greeting, reason for leave, requested date, assurance about missed assignments, and a respectful closing. Use formal English.

Output:

Dear Sir/Madam,

I am writing to inform you that I am feeling unwell and unable to attend college. I kindly request leave for one day, [date], to rest and recover.

I will ensure that I complete any missed assignments and catch up on the lessons covered during my absence.

I kindly request you to approve my leave application. Thank you for your understanding and consideration.

Yours sincerely,
[Your Name]
[Department and Class]

2. One-shot Prompt

Prompt:

Example:

Subject: Leave Request for Personal Reasons

Dear Sir/Madam,

I am unable to attend college due to a personal commitment. I kindly request leave for [date]. I will complete any missed academic work.

Thank you for your understanding.

Yours sincerely,
[Your Name]

Now write a similar professional email requesting leave due to illness. Include the date, a polite request, and a commitment to complete missed work.

Output:

Dear Sir/Madam,

I am writing to inform you that I am suffering from illness and will be unable to attend college on [date]. Therefore, I kindly request you to grant me leave for the day.

I will make sure to catch up on the lessons and complete any assignments that I miss during my absence.

I would be grateful for your approval. Thank you for your time and consideration.

Yours sincerely,
[Your Name]
[Department and Class]

3. Few-shot Prompt

Prompt:

Example 1:

Subject: Request for One-Day Leave

Dear Professor,

I am unable to attend class due to a personal reason. I kindly request leave for [date]. I will complete the pending coursework after my return.

Thank you for your consideration.

Yours sincerely,
[Student Name]

Example 2:

Subject: Leave Request for Medical Appointment

Dear Sir/Madam,

I have a medical appointment on [date] and will be unable to attend college. I kindly request leave for that day and will catch up on the missed lessons.

Thank you for your understanding.

Yours sincerely,
[Student Name]

Now write a professional email requesting leave due to illness. Follow the examples and include a subject, reason, dates, academic responsibilities, and polite closing.

Output:

Dear Sir/Madam,

I am writing to inform you that I am unwell and unable to attend college. I kindly request medical leave from [start date] to [end date] to rest and recover.

I understand the importance of attending classes and completing academic work. I will coordinate with my classmates and faculty to catch up on missed lessons and complete any pending assignments.

I would be grateful if you could approve my leave request for the specified period. Thank you for your understanding and consideration.

Yours sincerely,
[Your Name]
[Department and Class]
[Register Number]

Comparison of generated emails

Criteria	Zero-shot	One-shot	Few-shot
Tone	Professional	Polite	Professional and considerate
Grammar	Correct	Correct	Correct
Formatting	Standard email	Standard email	Detailed email structure
Completeness	Good	Good	Very good

Result: Few-shot prompting encourages a complete email by demonstrating how to include dates, academic responsibilities, and a formal closing.

Q5. Promotional Social Media Post for an AI Workshop

Aim: To design Zero-shot, One-shot, and Few-shot prompts to generate a promotional social media post for an AI Workshop.

Algorithm:

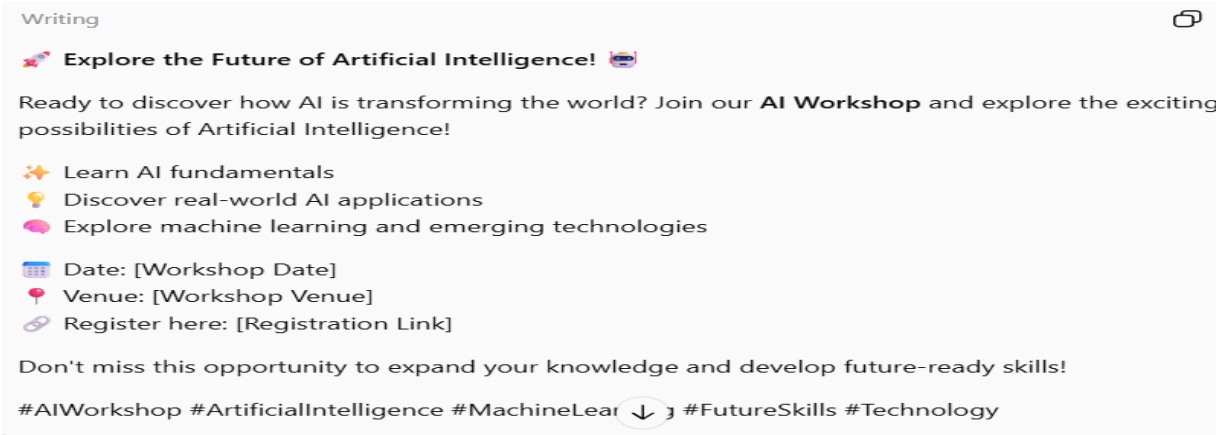
1. Start the program.
2. Define the task of promoting an Artificial Intelligence Workshop.
3. Create a Zero-shot prompt without examples.
4. Create a One-shot prompt with one sample promotional post.
5. Create a Few-shot prompt with multiple sample promotional posts.
6. Generate a promotional post using each prompting method.
7. Display the generated posts.
8. Compare the posts based on creativity, clarity, structure, and engagement.
9. Stop the program.

1. Zero-shot Prompt

Prompt:

Create an engaging social media promotional post for an Artificial Intelligence Workshop. Include an attractive headline, workshop benefits, date, venue, registration details, a call to action, and relevant hashtags. Use an enthusiastic tone.

Output:



2. One-shot Prompt

Prompt:

Example:

🎯 Join Our Python Workshop!

Learn programming fundamentals, develop coding skills, and explore practical applications through an engaging workshop.

🗓️ Date: [Date]
📍 Venue: [Venue]
🔗 Register now: [Link]

#PythonWorkshop #Coding #Technology

Now create a similar promotional social media post for an AI Workshop. Include learning benefits, event details, a call to action, emojis, and relevant hashtags.

Output:

Writing

🤖 **Unlock the Power of Artificial Intelligence!**

Join our exciting **AI Workshop** and discover how Artificial Intelligence is changing industries, education, healthcare, and everyday life.

✨ Understand AI fundamentals
💻 Explore practical AI applications
🚀 Build your knowledge of future technologies

🗓️ Date: [Workshop Date]
📍 Venue: [Workshop Venue]
🔗 Register today: [Registration Link]

Take the next step toward your AI learning journey. We look forward to seeing you there!

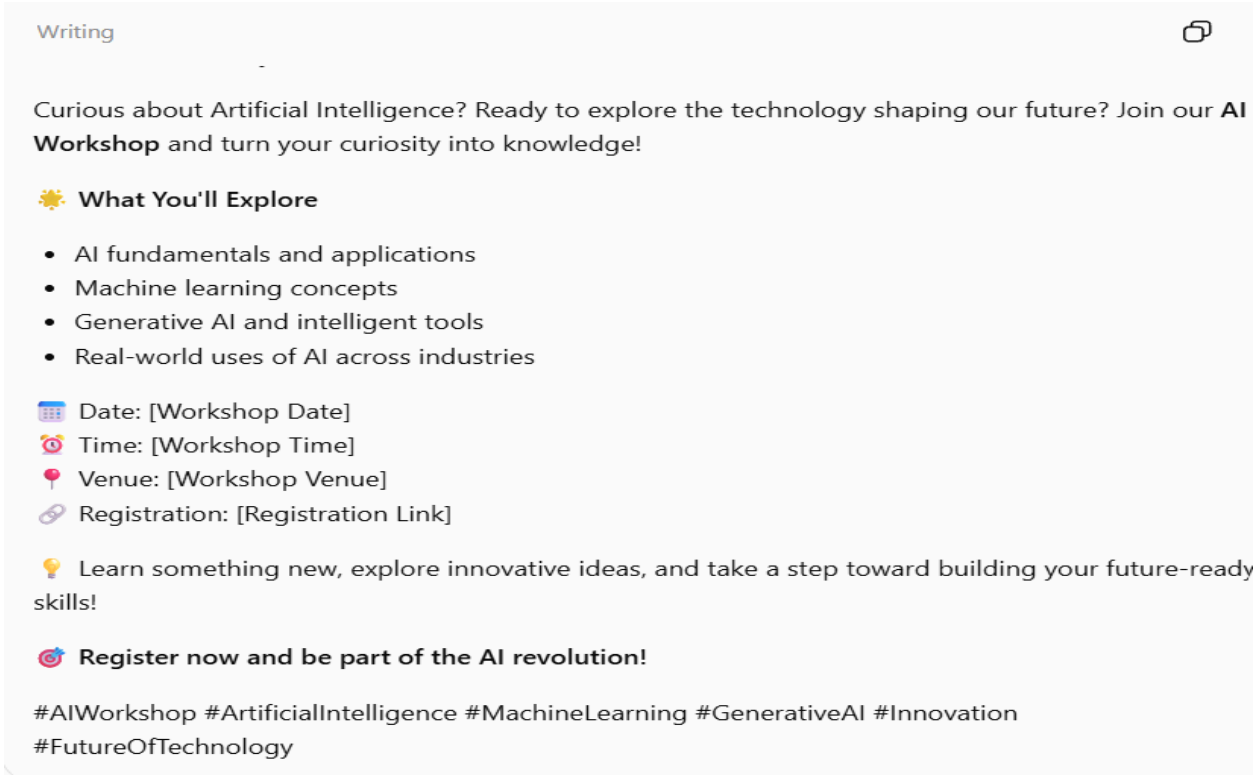
#AIWorkshop #ArtificialIntelligence #GenerativeAI #Innovation #DigitalSkills

3. Few-shot Prompt

Prompt:



Output:



Result: Few-shot prompting generates a structured and engaging promotional post by learning the format and style demonstrated in multiple examples.

Q6: Text Generation Using OpenAI API

Aim: To develop a Python program that connects to an OpenAI Large Language Model (LLM) using the OpenAI API and generates text based on a user-provided prompt.

Algorithm

1. Start the program.
2. Install and import the OpenAI library.
3. Enter the OpenAI API key securely.
4. Create an OpenAI client.
5. Get a text prompt from the user.
6. Send the prompt to the OpenAI model.
7. Display the generated text.
8. Stop the program.

```python
from transformers import pipeline

# Load a free, pre-trained text-generation model
generator = pipeline(
    "text-generation",
    model="distilgpt2"
)

# Get input from the user
prompt = input("Enter your prompt: ")

# Generate text
result = generator(
    prompt,
    max_new_tokens=80,
    do_sample=True,
    temperature=0.7,
    pad_token_id=generator.tokenizer.eos_token_id
)

# Display generated text
print("\nGenerated Text:")
print(result[0]["generated_text"])
```

```
Loading weights: 100% 76/76 [00:00<00:00, 2904.96it/s]
Enter your prompt: Explain Artificial Intelligence in simple words

Generated Text:
Explain Artificial Intelligence in simple words, and I think you'll love it.


This post is about the artificial intelligence and the artificial intelligence.
It's about the potential for artificial intelligence to be made better and more effective by the use of artificial intelligence.
The AI could be a good thing for us to understand more about human behavior. (That's why I'm writing this post about artificial intelligence.)
In fact
```

Result

Thus, a Python program was developed to connect to the OpenAI API and generate text based on a user-provided prompt.

Q7: Text Generation Using Gemini API

Aim: To develop a Python program that connects to the Google Gemini API and generates text based on a user-provided prompt.

Algorithm

1. Start the program.
2. Install and import the Google GenAI library.
3. Enter the Gemini API key.
4. Create the Gemini client.
5. Get a prompt from the user.
6. Send the prompt to the Gemini model.
7. Display the generated text.
8. Stop the program.

code:

```
from transformers import pipeline

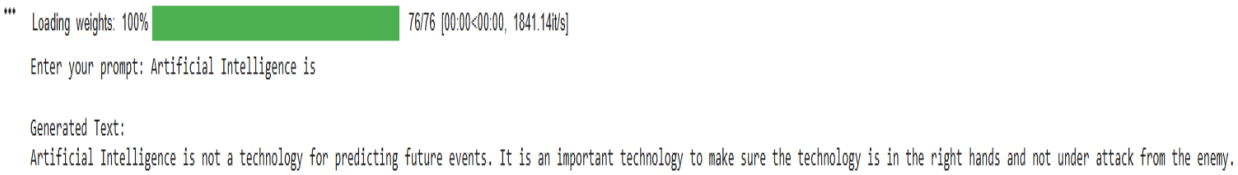
# Load a pre-trained language model
generator = pipeline(
    "text-generation",
    model="distilgpt2"
)

# Get prompt from user
prompt = input("Enter your prompt: ")

# Generate text
result = generator(
    prompt,
    max_new_tokens=60,
    do_sample=True,
    temperature=0.7,
    pad_token_id=generator.tokenizer.eos_token_id
)

# Display output
print("\nGenerated Text:")
print(result[0]["generated_text"])
```

OUTPUT:



Result: Text generation was performed using a pre-trained language model in Google Colab without requiring an API key.

8: Text Generation Using Hugging Face Inference API

Aim: To develop a Python program that uses the Hugging Face Inference API to generate text from a user-provided prompt.

Algorithm

1. Start the program.
2. Install the Transformers and PyTorch libraries.
3. Load a pre-trained text-generation model.
4. Get a prompt from the user.
5. Generate text using the model.
6. Display the generated text.
7. Stop the program.

PROGRAM

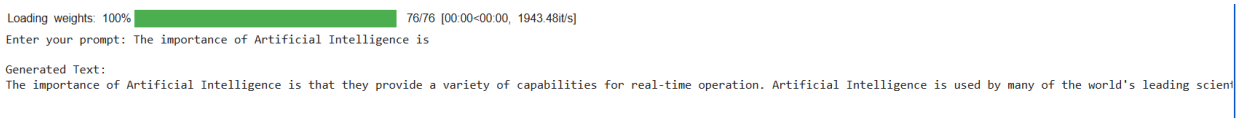
```python
from transformers import pipeline

# Load a pre-trained Hugging Face model
generator = pipeline(
    "text-generation",
    model="distilgpt2"
)

# Get input from the user
prompt = input("Enter your prompt: ")

# Generate text
result = generator(
    prompt,
    max_new_tokens=80,
    do_sample=True,
    temperature=0.7,
    pad_token_id=generator.tokenizer.eos_token_id
)

# Display the result
print("\nGenerated Text:")
print(result[0]["generated_text"])
```



Result

Thus, text generation was performed using a pre-trained Hugging Face model in Google Colab without requiring an API key

Q9: Python Code Generation Using Structured Prompting

Aim: To generate Python code using a structured prompt and verify its correctness by executing test cases.

Algorithm

1. Start the program.
2. Define a structured prompt containing the task, input, output, and constraints.
3. Generate Python code for calculating the factorial of a number.
4. Execute the generated code.
5. Test the code using sample inputs.
6. Display the output and validation results.
7. Stop the program.

PROGRAM:

```
# Q9: Python Code Generation Using Structured Prompting

# Step 1: Define a structured prompt
prompt = """
Task: Write a Python program to calculate factorial.
Input: A non-negative integer.
Output: Factorial of the given number.
Constraint: Negative numbers must raise ValueError.
"""

print("Structured Prompt:")
print(prompt)

# Step 2: Python code generated for the task
def factorial(n):
    if n < 0:
        raise ValueError("Number must be non-negative")

    result = 1
    for i in range(1, n + 1):
        result *= i

    return result

# Step 3: Test the code
print("\nTest Results:")
print("Factorial of 0:", factorial(0))
print("Factorial of 3:", factorial(3))
print("Factorial of 5:", factorial(5))

# Step 4: Validate the results
assert factorial(0) == 1
assert factorial(3) == 6
assert factorial(5) == 120

try:
    factorial(-2)
except ValueError:
    print("Negative input test: Passed")

print("All test cases passed!")
```

OUTPUT:

```
Structured Prompt:

Task: Write a Python program to calculate factorial.
Input: A non-negative integer.
Output: Factorial of the given number.
Constraint: Negative numbers must raise ValueError.


Test Results:
Factorial of 0: 1
Factorial of 3: 6
Factorial of 5: 120
Negative input test: Passed
All test cases passed!
```

Result

Thus, structured prompting was demonstrated by defining clear requirements, implementing the corresponding Python code, and validating its output with test cases.

Q10. SQL Query Generation Using Structured Prompting

Aim: To generate an SQL query using a structured prompt and verify its output.

Algorithm:

1. Define the student table structure.
2. Create a structured prompt describing the SQL task.
3. Write an SQL query to select students scoring above 80.
4. Sort the results in descending order of marks.
5. Execute the query and verify the output.

Program:

```
import sqlite3

# Create an in-memory database
conn = sqlite3.connect(":memory:")
cursor = conn.cursor()

# Create Students table
cursor.execute("""
CREATE TABLE Students (
    student_id INTEGER PRIMARY KEY,
    name TEXT,
    department TEXT,
    marks INTEGER
)
""")

# Insert sample records
students = [
    (1, "Arun", "CSE", 90),
    (2, "Priya", "IT", 85),
    (3, "Kumar", "ECE", 70),
    (4, "Divya", "CSE", 95)
]
cursor.executemany("INSERT INTO Students VALUES (?, ?, ?, ?)", students)

# Structured prompt
prompt = """
Task: Select student names and marks.
Condition: Marks must be greater than 80.
Sorting: Marks in descending order.
"""

print("Structured Prompt:")
print(prompt)
```

```python
print("Structured Prompt:")
print(prompt)

# SQL query
query = """
SELECT name, marks
FROM Students
WHERE marks > 80
ORDER BY marks DESC;
"""

print("Generated SQL Query:")
print(query)

# Execute and display results
cursor.execute(query)
results = cursor.fetchall()

print("Query Output:")
for row in results:
    print(row)

# Validate results
assert all(marks > 80 for name, marks in results)
assert results == sorted(results, key=lambda row: row[1], reverse=True)

print("Query validation successful!")

conn.close()
```

Output:

```
Structured Prompt:

Task: Select student names and marks.
Condition: Marks must be greater than 80.
Sorting: Marks in descending order.

Generated SQL Query:

SELECT name, marks
FROM Students
WHERE marks > 80
ORDER BY marks DESC;

Query Output:
('Divya', 95)
('Arun', 90)
('Priya', 85)
Query validation successful!
```

Result: The SQL query was executed successfully and returned students who scored above 80, sorted by marks in descending order.

Q10. SQL Query Generation Using Structured Prompting

Aim: To generate an SQL query using a structured prompt and verify its output.

Algorithm:

1. Define the student table structure.
2. Create a structured prompt describing the SQL task.
3. Write an SQL query to select students scoring above 80.
4. Sort the results in descending order of marks.
5. Execute the query and verify the output.

Program:

```
import sqlite3

# Create an in-memory database
conn = sqlite3.connect(":memory:")
cursor = conn.cursor()

# Create Students table
cursor.execute("""
CREATE TABLE Students (
    student_id INTEGER PRIMARY KEY,
    name TEXT,
    department TEXT,
    marks INTEGER
)
""")

# Insert sample records
students = [
    (1, "Arun", "CSE", 90),
    (2, "Priya", "IT", 85),
    (3, "Kumar", "ECE", 70),
    (4, "Divya", "CSE", 95)
]
cursor.executemany("INSERT INTO Students VALUES (?, ?, ?, ?)", students)

# Structured prompt
prompt = """
Task: Select student names and marks.
Condition: Marks must be greater than 80.
Sorting: Marks in descending order.
"""

print("Structured Prompt:")
print(prompt)
```

```python
print("Structured Prompt:")
print(prompt)

# SQL query
query = """
SELECT name, marks
FROM Students
WHERE marks > 80
ORDER BY marks DESC;
"""

print("Generated SQL Query:")
print(query)

# Execute and display results
cursor.execute(query)
results = cursor.fetchall()

print("Query Output:")
for row in results:
    print(row)

# Validate results
assert all(marks > 80 for name, marks in results)
assert results == sorted(results, key=lambda row: row[1], reverse=True)

print("Query validation successful!")

conn.close()
```

Output:

```
Structured Prompt:

Task: Select student names and marks.
Condition: Marks must be greater than 80.
Sorting: Marks in descending order.

Generated SQL Query:

SELECT name, marks
FROM Students
WHERE marks > 80
ORDER BY marks DESC;

Query Output:
('Divya', 95)
('Arun', 90)
('Priya', 85)
Query validation successful!
```

Result: The SQL query was executed successfully and returned students who scored above 80, sorted by marks in descending order.

Q12. Prompt Evaluation, Comparison, and Refinement

Aim: To evaluate different prompts, compare their responses, and refine the prompt to improve output quality.

Algorithm:

1. Define a task about the benefits of solar energy.
2. Create three prompts with different levels of detail.
3. Prepare sample responses for each prompt.
4. Evaluate the responses using relevance, accuracy, completeness, clarity, and format.
5. Calculate the average score and select the best refined prompt.

Program:

```
task = "Explain the benefits of solar energy."

prompts = {
    "Prompt A": "Explain solar energy.",
    "Prompt B": (
        "Explain five benefits of solar energy in simple English "
        "using bullet points."
    ),
    "Prompt C": (
        "Explain five benefits of solar energy in simple English. "
        "Use numbered points, include a short example, and conclude "
        "with one sentence about environmental protection."
    )
}

# Sample responses for demonstration
responses = {
    "Prompt A": (
        "Solar energy is renewable and helps reduce electricity costs."
    ),
    "Prompt B": (
        "1. Renewable source.\n"
        "2. Reduces electricity bills.\n"
        "3. Lowers air pollution.\n"
        "4. Requires little maintenance.\n"
        "5. Reduces dependence on fossil fuels."
    ),
    "Prompt C": (
        "1. Renewable: Sunlight is naturally available.\n"
        "2. Saves money: Solar panels can reduce electricity bills.\n"
        "3. Reduces pollution: Solar power produces no direct air "
        "pollution during operation.\n"
        "4. Low maintenance: Panels generally need limited upkeep.\n"
        "5. Energy independence: Solar power can reduce reliance "
```

```
        "Example: A home can use rooftop panels to generate electricit
        "Conclusion: Solar energy helps protect the environment."
    )
}

# Illustrative scores out of 5
scores = {
    "Prompt A": [4, 5, 2, 4, 2],
    "Prompt B": [5, 5, 4, 5, 5],
    "Prompt C": [5, 5, 5, 5, 5]
}

criteria = [
    "Relevance", "Accuracy", "Completeness",
    "Clarity", "Format Adherence"
]

for name in prompts:
    average = sum(scores[name]) / len(scores[name])
    print("\n", name)
    print("Prompt:", prompts[name])
    print("Response:", responses[name])
    print("Scores:", dict(zip(criteria, scores[name])))
    print("Average:", round(average, 2), "/ 5")

best = max(scores, key=lambda name:
           sum(scores[name]) / len(scores[name]))

print("\nBest Refined Prompt:", best)
print("Final Score:",
      round(sum(scores[best]) / len(scores[best]), 2), "/ 5")
print("Prompt evaluation completed successfully!")
```

Output:

```
...
 Prompt A
Prompt: Explain solar energy.
Response: Solar energy is renewable and helps reduce electricity costs.
Scores: {'Relevance': 4, 'Accuracy': 5, 'Completeness': 2, 'Clarity': 4, 'Format Adherence': 2}
Average: 3.4 / 5

 Prompt B
Prompt: Explain five benefits of solar energy in simple English using bullet points.
Response: 1. Renewable source.
2. Reduces electricity bills.
3. Lowers air pollution.
4. Requires little maintenance.
5. Reduces dependence on fossil fuels.
Scores: {'Relevance': 5, 'Accuracy': 5, 'Completeness': 4, 'Clarity': 5, 'Format Adherence': 5}
Average: 4.8 / 5

 Prompt C
Prompt: Explain five benefits of solar energy in simple English. Use numbered points, include a short example, and conclude with one sentence about environmental protection.
Response: 1. Renewable: Sunlight is naturally available.
2. Saves money: Solar panels can reduce electricity bills.
3. Reduces pollution: Solar power produces no direct air pollution during operation.
4. Low maintenance: Panels generally need limited upkeep.
5. Energy independence: Solar power can reduce reliance on grid electricity.
Example: A home can use rooftop panels to generate electricity.
Conclusion: Solar energy helps protect the environment.
Scores: {'Relevance': 5, 'Accuracy': 5, 'Completeness': 5, 'Clarity': 5, 'Format Adherence': 5}
Average: 5.0 / 5

Best Refined Prompt: Prompt C
Final Score: 5.0 / 5
Prompt evaluation completed successfully!
```

Result: Prompt C achieved the highest illustrative score because it provides clear instructions about content, structure, examples, and conclusion.